CSE-2103

OBJECT-ORIENTED PROGRAMMING

BASIC CONCEPT OF OBJECT-ORIENTED PROGRAMMING (OOP)

Conducted by

Mehedi Hasan

Lecturer, Department of CSE

WHAT IS OBJECT-ORIENTED PROGRAMMING (OOP)

- OOP is a programming paradigm based on the concept of 'objects'.
- Objects contain data in the form of attributes and code in the form of methods.
- It helps model real-world systems using classes and objects, improving modularity, reusability, and maintainability.

KEY FEATURES OF OOP

- 1. Class: Blueprint for creating objects.
- 2. Object: Instance of a class.
- 3. Encapsulation: Binding data and methods.
- 4. Inheritance: Reuse properties of existing classes.
- 5. Polymorphism: One interface, many implementations.
- 6. Abstraction: Hiding complex details from the user.

WHAT IS STRUCTURE-ORIENTED PROGRAMMING (SOP)

- SOP focuses on procedures or routines (functions).
- It separates data and functions.

- Programs are organized as a set of functions and global data.

COMPARISON: OOP VS SOP

Aspect	OOP	SOP
Approach	Object-based	Function-based
Modularity	High (Classes & Objects)	Low (Functions only)
Data Security	Supports encapsulation	No encapsulation
Code Reusability	Via inheritance	Limited reuse
Real-world Modeling	Easy	Difficult

OBJECT ORIENTED PROGRAMMING (OOP)

Real-World Entity:

- ✓ In Real-World every single entity is unique, like…….
- ✓ Every object have some attributes. (Data)
- ✓ Also, performs some actions. (Functions)

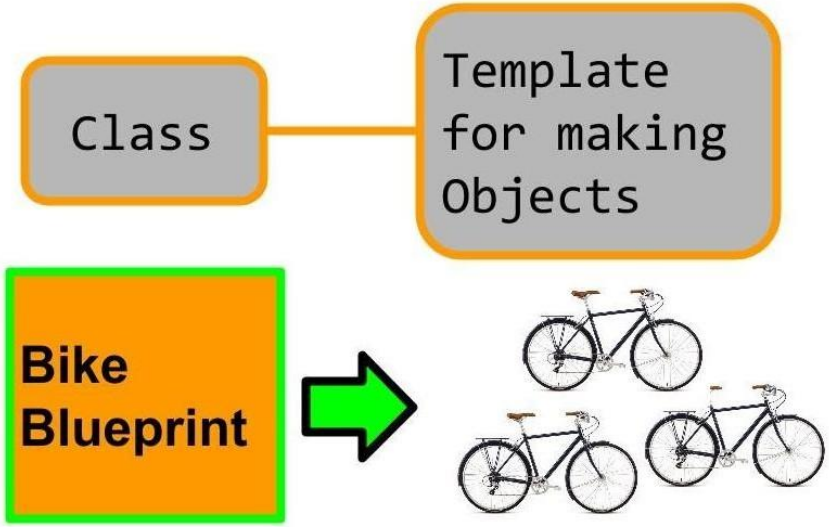
Object:

- ✓ The fundamental idea of OOP is to combine both data and functions into a single unit.
- ✓ Such unit is called an object.
- ✓ The functions of an object operates on these data.

OBJECT ORIENTED PROGRAMMING

Class:

- ✓ Class is the building block of OOP.
- ✓ It is a user-defined data type, which holds its own **data members and member functions**, which can be accessed and used by creating an instance of that class.
- ✓ A class is like a **blueprint** for an object.
- ✓ An Object is an **instance** of a Class.
- ✓ When a class is defined, no memory is allocated but when it is instantiated (i.e. an object is created) memory is allocated.



OBJECT ORIENTED PROGRAMMING (OOP)

Data Encapsulation:

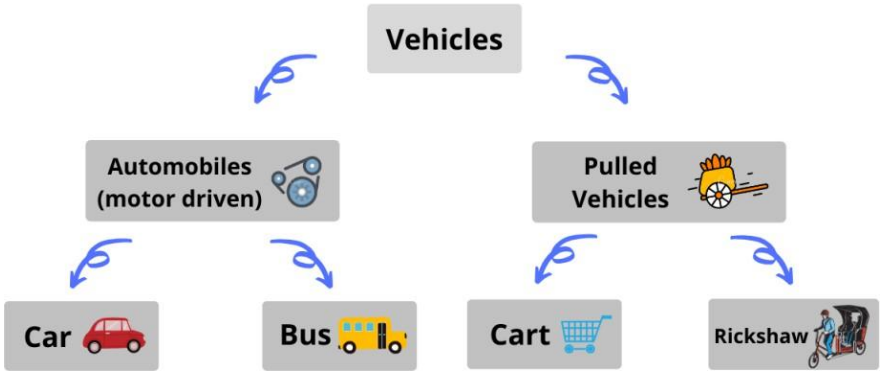
- ✓ Binding (or wrapping) code and data together into a single unit is known as encapsulation.
- ✓ Data encapsulation is a key term of OOP.
- ✓ Outside world can't access data of an object directly.
- ✓ Data of an Object only can be accessed by using its member function.
- ✓ Data is hidden, so it is safe from accidental use and alteration.

OBJECT ORIENTED PROGRAMMING (OOP)

Inheritance:

- ✓ Inheritance is one of the most important feature of Object Oriented Programming.
- ✓ It allows to create a **new class** (derived class) from an **existing class** (base class).
- ✓ The derived class gets **all the features** from the base class and can have **additional features.**
- ✓ Inheritance is an **is-a relationship**. We use inheritance only if an is-a relationship is present between the two classes.

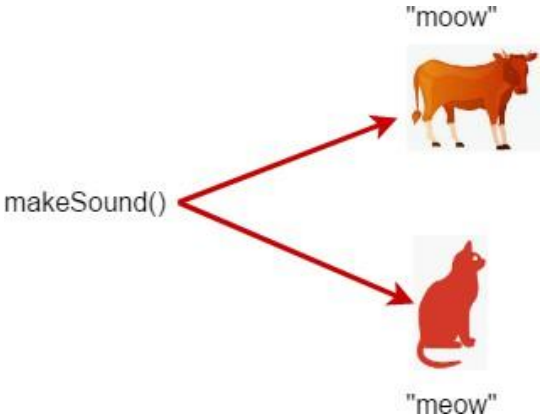
- A car is a vehicle. - Orange is a fruit. - A surgeon is a doctor. - A dog is an animal.



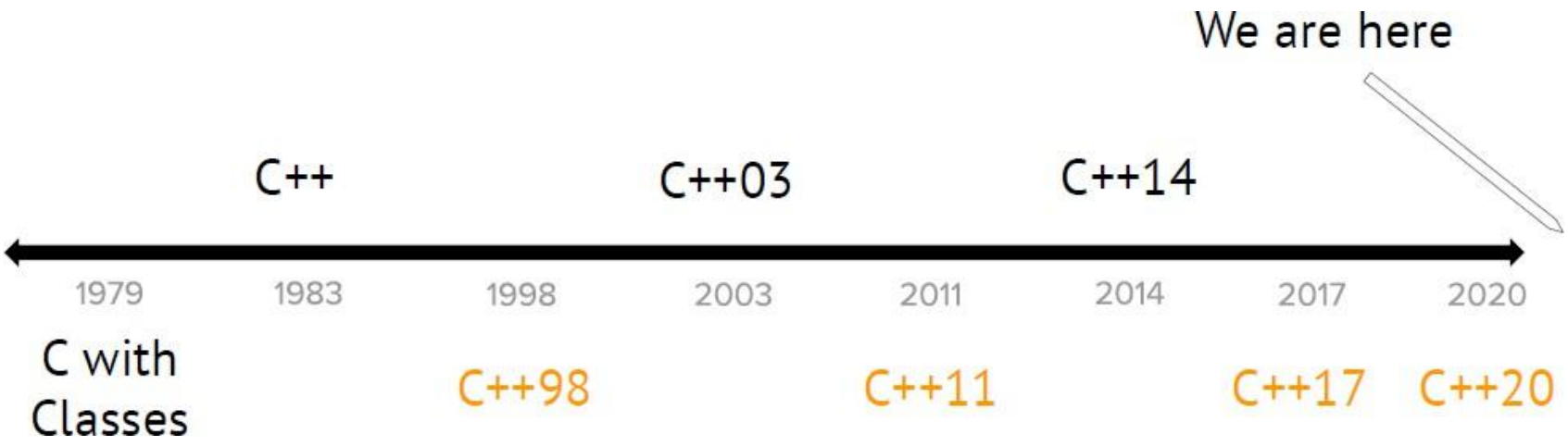
OBJECT ORIENTED PROGRAMMING (OOP)

Polymorphism and Overloading:

- ✓ Polymorphism means many forms.
- ✓ In C++, polymorphism causes a member function to behave differently based on the object that calls/invokes it.



C++ HISTORY: EVOLUTION OF C++



BASIC SYNTAX OF C++

```
#include <iostream>          // Preprocessor directive
using namespace std;         // Use standard namespace

int main() {                 // Main function
    cout << "Hello, World!"; // Output statement
    return 0;                // End of program
}
```

PROGRAM SYNTAX

C Programming

```c
#include <stdio.h>

int main() {
    printf("Hello, world\n");
    return 0;
}
```

C++ Programming

```cpp
#include <iostream>
using namespace std;

int main()
{
    cout << "Hello, World!\n";
    return 0;
}
```

CONSOLE I/O

C Programming

```
printf("Text to display.");

scanf("%d", &a);
```

C++ Programming

```
cout << "Text to display.";

cin >> a;
```

EXPLANATION OF C++ SYNTAX

- #include <iostream>: Includes the standard input-output stream library.
- using namespace std;: Avoids writing 'std::' before standard functions.
- int main(): Entry point of every C++ program.
- cout: Used to print output to the console.
- return 0;: Indicates that the program ended successfully.

SUMMARY

- OOP helps in building scalable and maintainable applications.
- It reflects real-world behavior more naturally than SOP.
- Encapsulation, inheritance, and polymorphism make OOP more powerful.
- SOP is simple but less suitable for complex systems.